SOKOINE UNIVERSITY OF AGRICULTURE

SCHOOL OF ENGINEERING AND TECHNOLOGY (SoET)

DEPARTMENT OF AGRICULTURAL ENGINEERING (DAE)

AE 219

ASSIGNMENT 2

Note:

[1] Please listen to videos https://www.youtube.com/playlist?list=PLmKnXyyknMkPhutqZ6uebPazrUCobJ8Ep provided by my Lab YEESI Lab and instructed by Dr Masamaki

[2] Please use Google Colab: https://colab.research.google.com/

[3] Please read and listen to the videos in https://towardsdatascience.com/best-time-saving-python-tricks-in-jupyter-notebook-fca877132507

Questions:

1. Write a Python program to calculate and display the sum of the series (1*1) + (1*2) + (1*2*3) + (1*2*3*4) + (1*2*3*4*5) + ... + (1*2*3*4*5….*n). Assume n=40. Validate **your program using Google Colab and submit it as a Jupyter notebook program in iPynb extension with comments.**
2. Write a Python program to calculate and display the sum of the series 1*2/1! + 2*3/2! + 3*4/3! + 4*5/4! +…….+ n*(n+1)/n!. Assume ! means factorial and n=40. **Validate your program using Google Colab and submit it as a Jupyter notebook program in iPynb extension with comments.**
3. Navigate and download the test dataset found in the following link: https://zindi.africa/competitions/tanzania-tourism-prediction/data
   Then, write a python program to calculate the correlation between the column total_males and total_females. **Validate your program using Google Colab and submit it as a Jupyter notebook program in iPynb extension with comments.**

   Submit your code results by Wednesday midnight (11-May-2022) to the CR as the softcopy and CR will forward the bundle of all codes to kadefue@sua.ac.tz. DO NOT SEND YOUR WORK to Instructor's email. Any enquiries, please send an email.

As usual, Every one hour late equates to 5 marks deduction.